01 Command Centre Dashboard

Sensor cards, live chart, risk ranking, alerts, forecast summary and system status in one glance.

FLOODGUARD AI MODE: LIVE · Connected · Model: RF v1.0.3 · 3 ALERTS · 14:32 IST

- Dashboard
- AI Prediction
- 2D Flood Map
- 3D Digital Twin
- What-If Simulator
- Alerts & Monitoring
- About This Data

Dark command-centre theme

Command Centre Dashboard

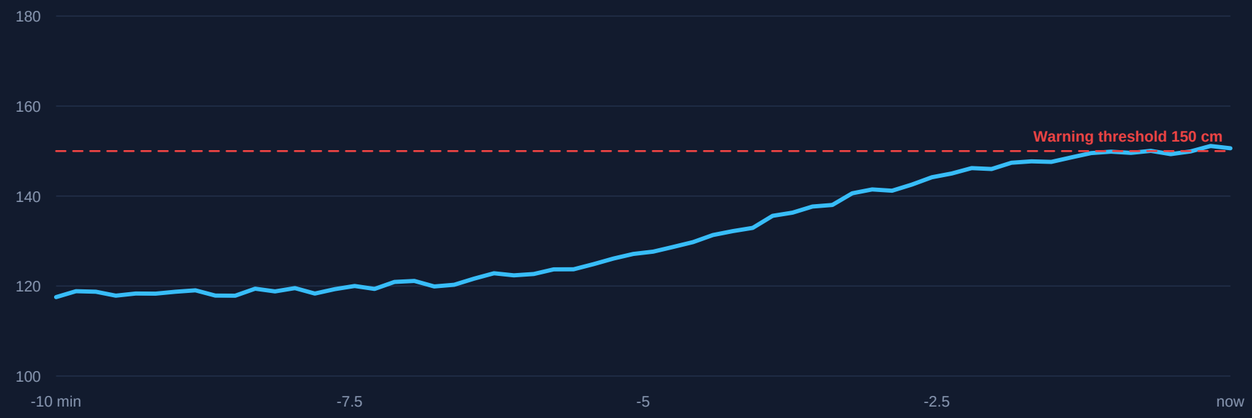
Mithi River corridor, BKC to Kurla

ACTIVE SENSORS	OFFLINE / STALE	RAINFALL NOW	HIGHEST RISK
4 / 5	1	4.2 mm/hr	HIGH

MR-P1	MR-V1	MR-V2	MR-V3	MR-V4
Channel (physical)	Upstream, Kurla	Midstream	Downstream, BKC	Mahim Bay mouth
LIVE HARDWARE	SIMULATED	SIMULATED	SIMULATED	SIMULATED
148 cm	122 cm	136 cm	157 cm	--
4.2 mm/hr	3.1 mm/hr	3.6 mm/hr	4.0 mm/hr	-- mm/hr
HIGH	LOW	MODERATE	HIGH	OFFLINE
12 s ago	8 s ago	8 s ago	8 s ago	STALE 6 min

LIVE HARDWARE

LIVE WATER LEVEL - MR-P1



HIGHEST-RISK LOCATIONS

MR-V3 Downstream, BKC	HIGH
MR-P1 Channel (physical)	HIGH
MR-V2 Midstream	MODERATE

LATEST ALERTS

HIGH	MR-P1 level above threshold
WARNING	MR-V3 forecast High in 30 min
INFO	MR-V4 stopped reporting

+15 min	+30 min	+60 min
157 cm	166 cm	181 cm
HIGH	HIGH	SEVERE

LEGEND

LOW · MODERATE · HIGH · SEVERE

LIVE HARDWARE · SIMULATED

Colour is always paired with a text label

Connection: Connected | Model: RF v1.0.3 | Data mode: Live | Last update: 14:32:08 IST

FLOODGUARD AI UI mockup. All values are sample data for illustration. Badges mark Live Hardware vs Simulated sources.

02 AI Prediction Page

History plus forecast with uncertainty band, explainable risk score, model card and baseline comparison.

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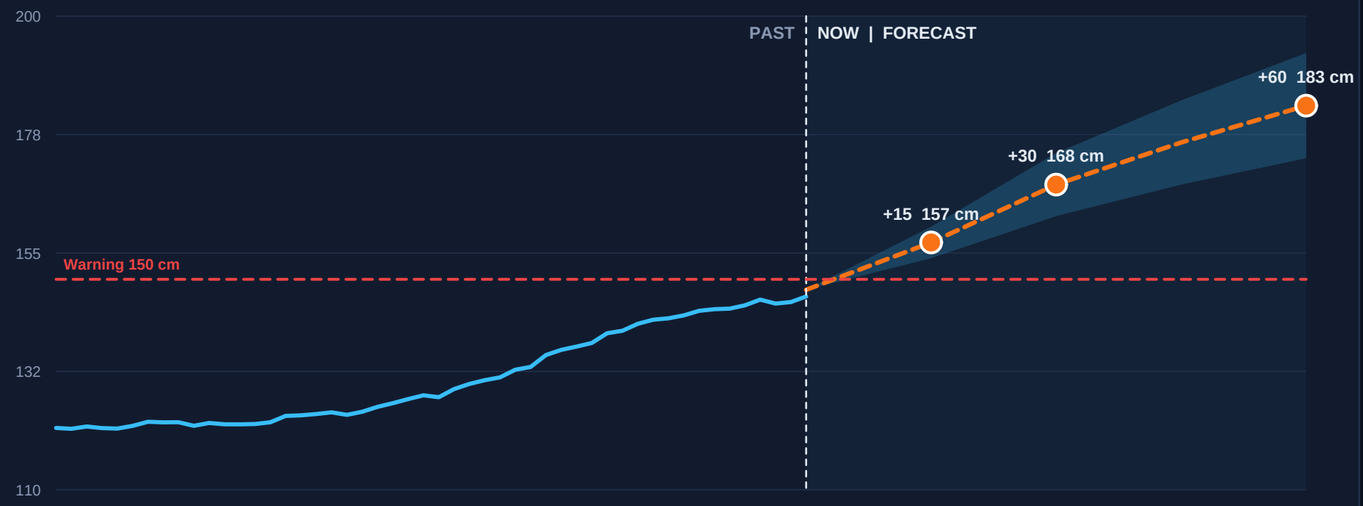
AI Prediction

Forecast with uncertainty and explainable risk

Node: MR-P1 Channel (physical) v

LIVE HARDWARE

WATER LEVEL: HISTORY AND FORECAST (cm)



CURRENT RISK SCORE

0.68 HIGH

Water level
Rise rate
Rainfall
Soil saturation

Driven mainly by rapid rise rate

MODEL

Random Forest Regressor, v1.0.3

Trained 04 Oct, 11:20 IST

Data: synthetic + prototype recordings

Horizons: 15, 30, 60 min

MODEL ERROR (MAE, cm) VS PERSISTENCE BASELINE - sample values



IN PLAIN WORDS

Water level at this node is forecast to rise over the next hour. Risk is expected to reach High within about 15 minutes and Severe by +60 minutes.

Input: live hardware readings
Forecast made at 14:32 IST

Error values are placeholders. A few minutes of prototype data cannot establish real-world accuracy.

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FLOODGUARD AI UI mockup. All values are sample data for illustration. Badges mark Live Hardware vs Simulated sources.

03 2D Flood Map

Sensors, flood depth layer and affected roads and buildings, driven by the shared forecast timeline.

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2D Flood Map

Leaflet + OpenStreetMap style view



LAYERS

- Sensors
- Flood heatmap
- Roads
- Buildings
- Flood boundaries

AFFECTED AT +30 MIN

Buildings	18
Road segments	5
Max depth	0.6 m

LEGEND

- Low depth
- Moderate depth
- High depth
- Severe depth
- Physical station
- Simulated station

Simplified elevation-based estimate. Not a hydraulic model. Accuracy limited by DEM resolution.

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04 3D Digital Twin

Terrain, extruded buildings, sensors and flood overlay, synchronised with the same timeline.

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3D Digital Twin

CesiumJS terrain, buildings and flood overlay

FORECAST +30 MIN

Affected buildings turn red



Now | +15 min | +30 min | +60 min | Play

CAMERA PRESETS

- Corridor overview
- Physical sensor close-up
- Along the river

SELECTED BUILDING

Building B-1042

AFFECTED

Estimated depth	0.4 m
Time step	+30 min

LAYERS

- Terrain
- Buildings
- Roads
- Flood overlay
- Sensors

Same flood layers as the 2D map. Falls back to local terrain, then to the 2D map if Cesium is unavailable.

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05 What-If Scenario Simulator

Sliders, presets, scenario banner, plain-language assumptions and result summary.

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What-If Scenario Simulator

Change conditions, see a different future

SCENARIO INPUTS

Scenario name: Heavy rain, drains at 70%

Rainfall intensity — 50 mm/hr

Initial water level — 150 cm

Drainage capacity — 70 %

Duration — 60 min

PRESETS

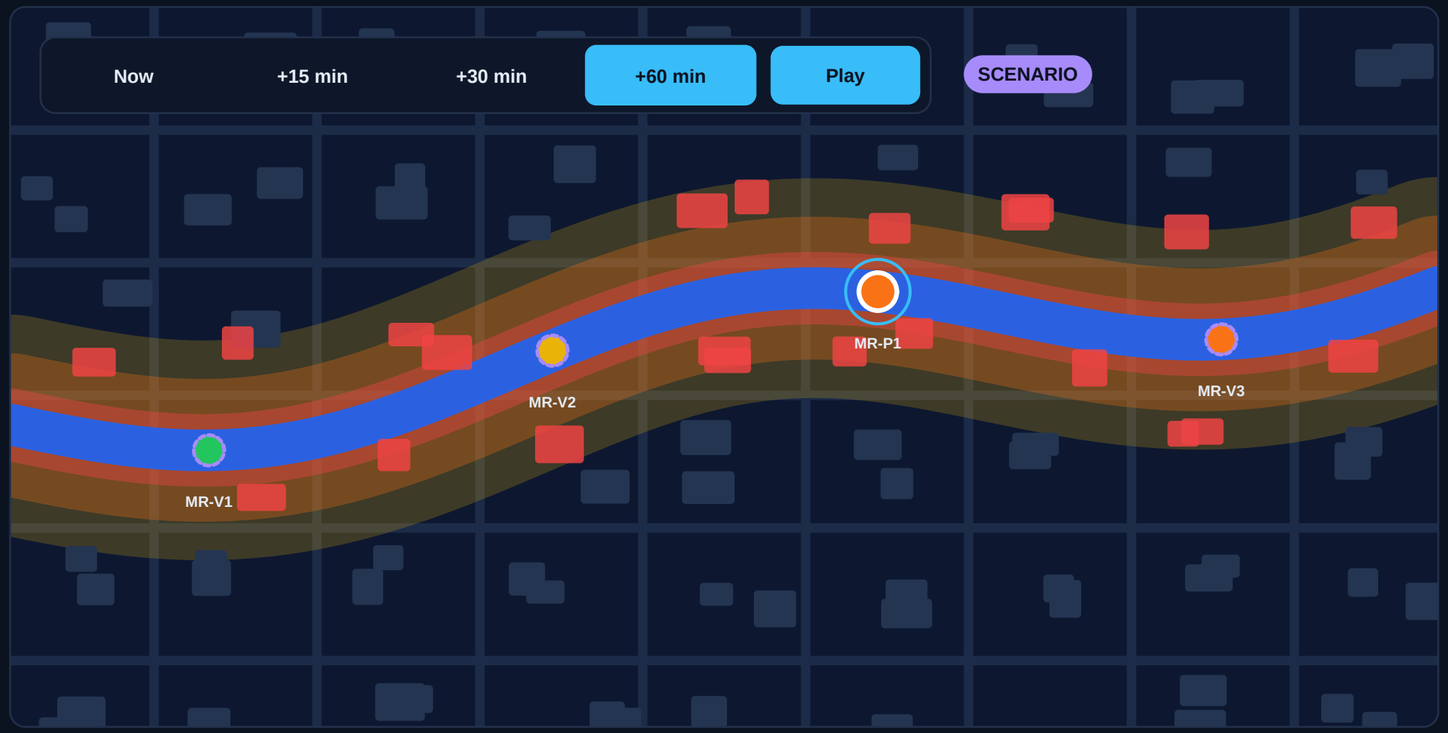
- Normal Monsoon Day
- Heavy Rain
- Extreme Rain, Poor Drainage

Run Scenario

Reset to live baseline

Estimated run time: a few seconds

SIMULATED WHAT-IF SCENARIO. You are not viewing live conditions.



ASSUMPTIONS USED

- Excess rain = rainfall x runoff coefficient (0.6), reduced by drainage capacity.
- Channel level rises by a set number of cm per mm of excess rain.
- Water spreads only to terrain lower than the water surface and connected to the river.
- Not a hydraulic model: no flow velocity. DEM is coarse (about 30 m).

SCENARIO RESULT

Peak level	Flooded area
192 cm	0.42 sq km

Roads affected	Buildings affected
9	31

Highest risk reached: SEVERE at +60

06 Scenario Comparison

Baseline versus scenario curves, split maps, difference numbers and an automatic summary sentence.

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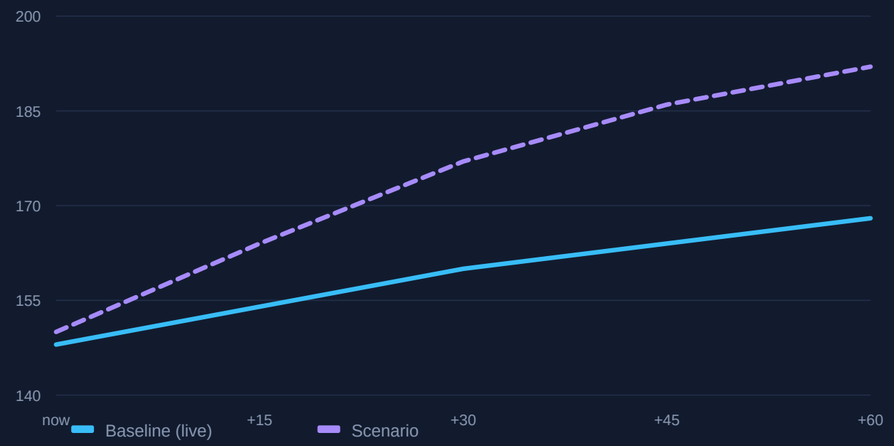
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Scenario Comparison

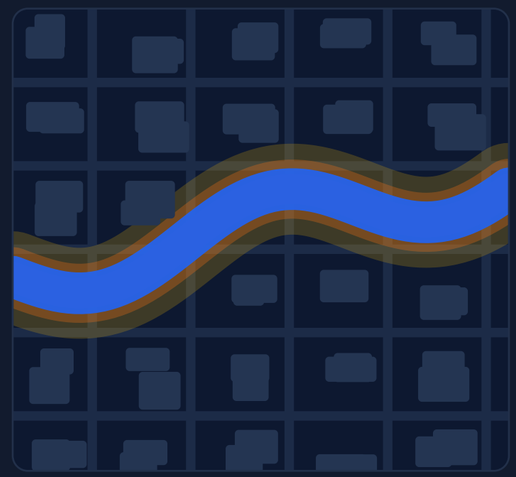
Baseline versus scenario

SIMULATED WHAT-IF SCENARIO: Heavy rain, drains at 70%. Baseline is live conditions.

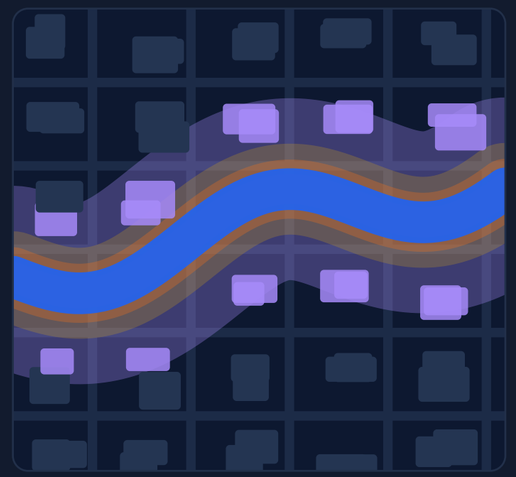
PREDICTED WATER LEVEL (cm)



Baseline



Scenario (extra area in violet)



PEAK LEVEL	FLOODED AREA	ADDITIONAL ROADS	ADDITIONAL BUILDINGS	HIGHEST RISK
192 cm	+0.27 sq km	+7	+14	SEVERE
baseline 168 cm	more than baseline	segments affected	by +60 min	baseline: High

Under the scenario, 14 additional buildings and 7 more road segments are affected by the +60 minute mark.

SAVED SCENARIOS

- Heavy rain, drains at 70% — SEVERE
- Normal monsoon day — MODERATE
- Extreme rain, poor drainage — SEVERE

TIMELINE (SHARED WITH MAPS)

Now | +15 min | +30 min | **+60 min** | Play

Showing predicted conditions at +60 minutes

FORECAST

07 Alerts and Monitoring

Active warnings, filterable history, network health and alert detail with acknowledgement.

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Alerts and Monitoring

Warnings, history and network health

CRITICAL — MR-P1 water level 152 cm is above the 150 cm warning threshold — Acknowledge

ACTIVE ALERTS

CRITICAL — 14:30 IST
Threshold - MR-P1
Level 152 cm above 150 cm threshold
Ack | Map

HIGH — 14:28 IST
Predicted risk - MR-V3
Forecast reaches Severe within 60 min
Ack | Map

WARNING — 14:26 IST
Comm. failure - MR-V4
No report for 6 minutes
Ack | Map

F

Node: All | Severity: All | Type: All | Last 24 h

14:30	CRITICAL	Threshold	MR-P1	Open
14:28	HIGH	Predicted risk	MR-V3	Open
14:26	WARNING	Comm. failure	MR-V4	Open
13:55	HIGH	Rapid rise	MR-P1	Acked 13:58
13:20	WARNING	Sensor fault	MR-P1	Cleared
12:48	INFO	Threshold	MR-V2	Cleared

NETWORK HEALTH

MR-P1	Online	12 s ago	4.9 V
MR-V1	Online	8 s ago	n/a
MR-V2	Online	8 s ago	n/a
MR-V3	Online	8 s ago	n/a
MR-V4	Offline	6 min ago	n/a

ALERT DETAIL

CRITICAL

Threshold alert - MR-P1

LIVE HARDWARE

Triggering readings



Created 14:30 IST | Not acknowledged

Acknowledge | View on map

Scenario alerts are tagged separately and never mixed with live ones.